

Bone Age Estimation from Hand Radiographs Using Lightweight CNNs with Patch-Based Fusion

Sobhan Hosseini

Department of Mathematics, University of Padova

sobhan.hosseini@studenti.unipd.it

Abstract—Bone age assessment from pediatric hand radiographs is a fundamental task in clinical practice, traditionally performed using atlas-based methods that are time-consuming and subject to inter-observer variability. In this course project, we investigate automated bone age estimation under constraints that require training convolutional neural networks entirely from scratch, without relying on pretrained backbones. To explicitly account for annotation uncertainty, we adopt an uncertainty-aware learning framework based on Deep Label Distribution Learning and evaluate multiple lightweight architectures, including whole-image models and patch-based variants with attention-based fusion. We further analyze the impact of incorporating gender metadata, increasing input resolution, and combining complementary models through ensembling. Experiments on the RSNA Pediatric Bone Age dataset show that gender information yields a substantial accuracy improvement with negligible computational overhead, while higher input resolution provides the largest single-model performance gain. Patch-based models achieve competitive accuracy at increased computational cost, highlighting a clear accuracy–complexity trade-off, and model ensembling provides additional robustness.

Index Terms—Bone age estimation, hand radiographs, convolutional neural networks, label distribution learning, efficiency-aware modeling

I. INTRODUCTION

Bone age assessment from pediatric hand radiographs is routinely used in clinical practice to estimate skeletal maturity and to support the diagnosis and monitoring of growth disorders. Conventional approaches, such as the Greulich–Pyle atlas, rely on expert visual comparison and are subject to inter-observer variability, motivating the development of automated, objective, and reproducible computational methods.

In this project, we address bone age estimation from hand X-ray images in a supervised learning setting, with performance evaluated primarily using mean absolute error (MAE) measured in months. Unlike many recent approaches that rely on large pretrained backbones and transfer learning, the ML4HD project constraints require training models *entirely from scratch*. Under this constraint, achieving competitive accuracy requires careful design choices in preprocessing, learning objectives, and model architecture, while explicitly accounting for computational efficiency.

We study bone age estimation through multiple complementary modeling strategies under identical experimental conditions, including a shared preprocessing pipeline, fixed dataset splits, and consistent optimization protocols. Specifically, we compare whole-image convolutional models with

patch-based architectures that aggregate local representations using attention-based fusion mechanisms. In both paradigms, we evaluate the impact of incorporating gender metadata as auxiliary information. Beyond architectural design, we also investigate the role of uncertainty-aware supervision via Deep Label Distribution Learning (DLDL) [1], the effect of increasing input resolution to preserve fine-grained anatomical detail, and the benefits of combining complementary models through ensembling.

Contributions.

- A medically motivated and deterministic preprocessing pipeline that suppresses acquisition artifacts and normalizes contrast and intensity statistics under a train-from-scratch setting.
- A whole-image convolutional baseline for bone age estimation trained entirely from scratch, with explicit analysis of accuracy–complexity trade-offs.
- Patch-based models with attention-based fusion mechanisms, enabling aggregation of local skeletal information and comparison between gated and learned fusion strategies.
- An uncertainty-aware learning framework based on Deep Label Distribution Learning, with expectation-based decoding for continuous age prediction.
- An empirical evaluation of gender metadata, input resolution scaling, and model ensembling, reporting both predictive performance (MAE in months) and efficiency metrics (parameter count and inference time).

The remainder of the paper is organized as follows. Section II reviews prior work on automated bone age assessment. Section III describes the overall processing pipeline. Section IV details the dataset and preprocessing steps. Section V presents the learning framework and model architectures. Section VI reports experimental results and analyses, and Section VII concludes with limitations and directions for future work.

II. RELATED WORK

Bone age estimation from pediatric hand radiographs has been extensively studied using deep learning methods. Early works demonstrated that convolutional neural networks (CNNs) can achieve clinically meaningful accuracy when trained on sufficiently large datasets. Larson et al. [2] showed that CNN-based approaches outperform traditional atlas-based

methods such as Greulich–Pyle, particularly when auxiliary metadata such as gender is incorporated.

The RSNA Pediatric Bone Age Machine Learning Challenge [3] established a standardized benchmark and highlighted the effectiveness of deep architectures combined with transfer learning and ensemble strategies. Top-performing submissions typically relied on large pretrained backbones (e.g., ResNet or Inception variants), extensive data augmentation, and multi-model ensembling, achieving low mean absolute error on the challenge test set.

While these approaches define strong upper-bound performance, their reliance on high-capacity pretrained models and substantial computational resources limits their applicability in constrained learning scenarios. In particular, training convolutional networks entirely from scratch—without external pretraining—remains comparatively underexplored, despite its relevance for educational settings, reproducible experimentation, and deployment contexts with limited access to large-scale pretraining corpora.

Several studies have also explored region-focused or patch-based strategies in medical imaging to emphasize anatomically informative areas and reduce background bias. Such approaches often depend on explicit anatomical localization, segmentation masks, or handcrafted priors. In contrast, we adopt a simple and fully automatic fixed overlapping patch extraction scheme combined with feature-level fusion, enabling a controlled comparison with whole-image models under identical preprocessing, optimization, and evaluation protocols.

In this work, we position our contribution at the intersection of these directions by systematically analyzing lightweight CNN architectures trained entirely from scratch, the impact of gender metadata, and the role of patch-based attention mechanisms. Unlike prior work focused on maximizing absolute performance through transfer learning, our study emphasizes controlled ablations and explicit accuracy–complexity trade-offs under fixed experimental constraints.

III. PROCESSING PIPELINE

This section provides a high-level overview of the end-to-end workflow, from input radiographs to bone age prediction. The pipeline is designed to enable a controlled comparison between whole-image and patch-based learning strategies under identical preprocessing and training conditions.

A. Whole-image prediction

In the whole-image setting, the complete hand radiograph is used as input. Each image is resized to a fixed resolution and intensity-normalized using statistics computed on the training set. The normalized image is processed by a lightweight convolutional neural network (CNN) encoder that extracts a global feature representation. This representation is mapped to a discrete probability distribution over age bins using a prediction head trained with the DLDL objective. When available, gender metadata is embedded and concatenated with the image-level feature vector prior to the final prediction layers.

TABLE I
DATASET STATISTICS FOR TRAINING AND VALIDATION SPLITS.

Statistic	Training	Validation
Number of images	12,611	1,425
Male / Female (count)	6,833 / 5,778	773 / 652
Male / Female (%)	54.18 / 45.82	54.25 / 45.75
Mean age (months)	127.32	127.16
Median age (months)	132	132
Std. deviation (months)	41.18	41.72
Age range (months)	1–228	3–228

B. Patch-based prediction with fusion

In the patch-based setting, the resized image is decomposed into a fixed grid of overlapping patches. Each patch is processed independently by the same CNN encoder used in the whole-image model, ensuring architectural consistency. This yields a set of patch-level feature embeddings, one per patch.

Patch embeddings are then aggregated by a fusion module to obtain a global representation used for age prediction. We consider two attention-based fusion strategies: (i) gated fusion, which modulates patch importance through a learnable gating mechanism, and (ii) learned softmax fusion, which assigns normalized data-driven weights to patches. Gender information, when used, is incorporated after patch aggregation by concatenation with the fused representation. This design ensures that whole-image and patch-based models differ only in input decomposition and fusion mechanism, enabling a fair and controlled comparison.

IV. DATA AND PREPROCESSING

A. Dataset

We use the pediatric hand radiograph dataset released for the RSNA Bone Age Machine Learning Challenge. The dataset consists of grayscale hand X-ray images annotated with skeletal age in months and binary gender metadata. Following the official dataset definition, we use the provided training set (12,611 images) for model learning and the validation set (1,425 images) for performance evaluation. The test set is not used, as ground-truth labels are not publicly available.

B. Dataset statistics

Table I reports summary statistics of bone age and gender distribution for the official splits. Figure 3 illustrates the distribution of samples across predefined age ranges for training and validation. The dataset is strongly imbalanced with respect to age, with substantially fewer samples in the youngest group (0–24 months). This imbalance is reflected in the error analysis across age ranges, as discussed in Section VI. For visualization purposes, ages above 216 months are assigned to the final bin.

C. Preprocessing

To reduce shortcut learning from non-anatomical cues and to improve robustness to acquisition variability, we apply a deterministic preprocessing pipeline to all images prior

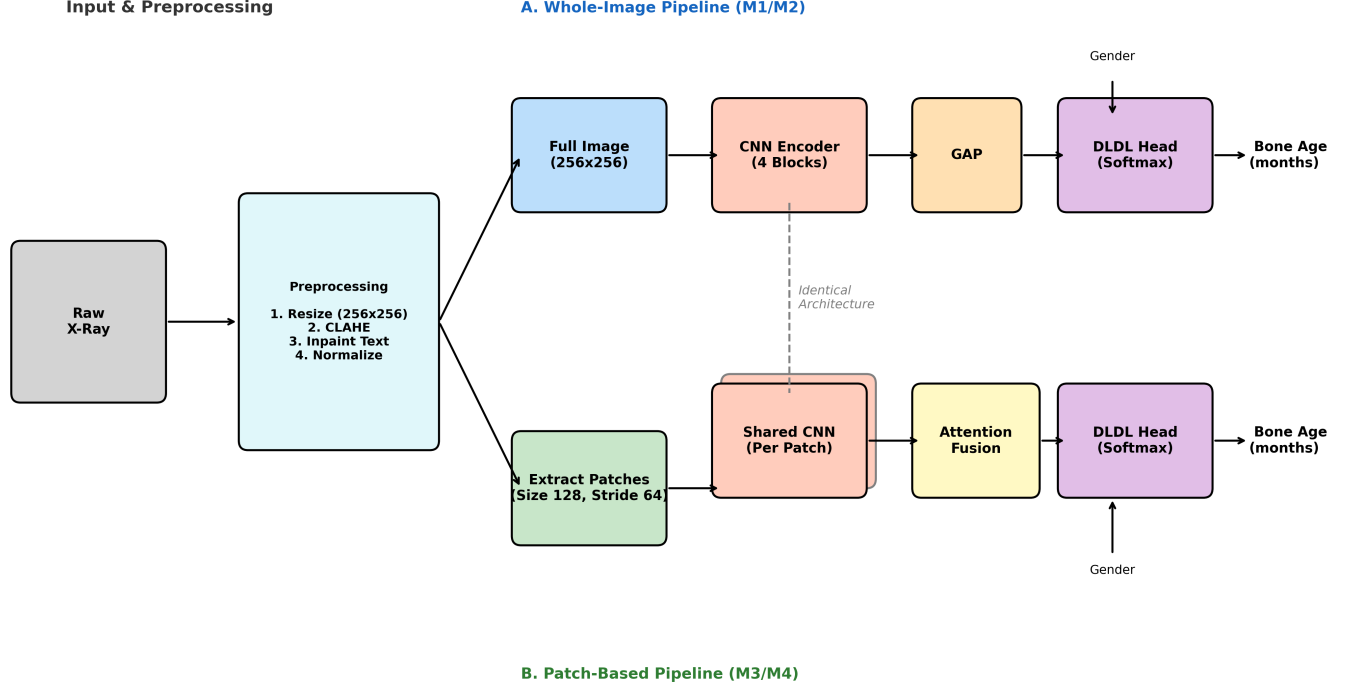


Fig. 1. Architectural overview of the Bone Age Estimation framework. All inputs undergo a shared preprocessing pipeline (left). **Top (A):** Whole-image approach where the full radiograph is encoded into a global descriptor. **Bottom (B):** Patch-based approach where the image is split into overlapping patches, processed by a weight-shared encoder, and aggregated via an attention mechanism. Both pathways use DLDL for uncertainty-aware prediction.



Fig. 2. Example patches extracted from a resized hand radiograph using a fixed overlapping grid (patch size 128×128 , stride 64). Some patches capture anatomically informative regions (e.g., wrist and phalanges), while others mainly correspond to background or low-information areas, motivating the use of attention-based fusion.

to model input. All preprocessing steps are identical across models and across training, validation, and inference phases; only data augmentation (Section IV-D) is restricted to the training set.

a) Resizing and intensity format.: Each radiograph is decoded as a single-channel image and resized to a fixed spatial resolution of 256×256 pixels using bilinear interpolation. Pixel values are represented in the $[0, 255]$ range prior to contrast processing.

b) Local contrast normalization (CLAHE).: Contrast-Limited Adaptive Histogram Equalization (CLAHE) is applied to normalize local contrast and enhance the visibility of skeletal edges in under- or over-exposed scans. We use a clip

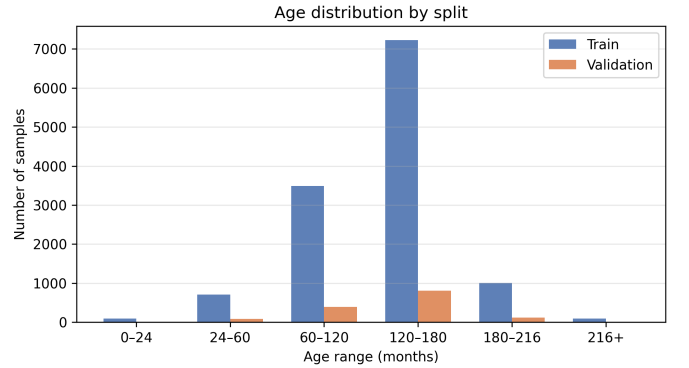


Fig. 3. Distribution of bone age across predefined age ranges for training and validation splits.

limit of 2.0 and an 8×8 tile grid.

c) Collimation artifact suppression.: Radiographs frequently contain dark collimation borders that are unrelated to anatomy and may bias learning. To suppress these artifacts, we mask a square border corresponding to 6% of the minimum image dimension on all sides by setting its intensity to zero.

d) Radiopaque annotation removal.: Radiopaque textual markers (e.g., “L/R”) can introduce spurious correlations with age or acquisition protocol. Candidate annotation pixels are detected using a high-intensity threshold at the 99.7th per-

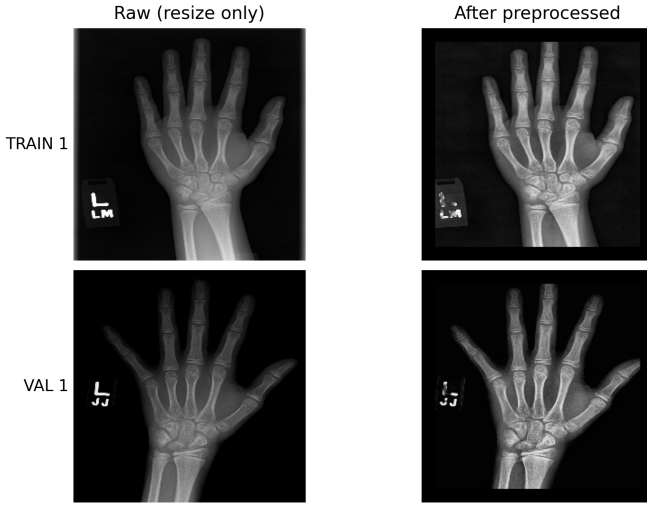


Fig. 4. Qualitative effect of preprocessing (raw image vs. after preprocessing). CLAHE enhances local contrast, while border masking and inpainting suppress acquisition-related artifacts.

centile of the preprocessed image. The resulting binary mask is refined using a 3×3 morphological opening and subsequently inpainted using the Telea method with radius 3.

e) Dataset-specific normalization with background exclusion.: Finally, inputs are normalized using a Z-score transformation

$$x_{\text{norm}} = \frac{x - \mu}{\sigma}. \quad (1)$$

To prevent the statistics from being dominated by the large black background, the mean μ and standard deviation σ are estimated on the *preprocessed training images only*, excluding near-background pixels (intensity > 10). This calibration is performed offline on a random subset of training samples, yielding $\mu = 68.8531$ and $\sigma = 55.3605$. These constants are fixed for all experiments to avoid information leakage and to ensure fair comparisons.

D. Data pipeline and augmentation

All experiments use a unified `tf.data` pipeline to ensure controlled and fair comparisons across architectures. Each sample consists of an image path, a binary gender indicator, and the ground-truth bone age in months.

a) Deterministic base mapping.: For both training and validation splits, the same deterministic preprocessing and normalization are applied. For whole-image models, the normalized image is provided directly as input. For patch-based models, a fixed grid of overlapping patches is extracted using `tf.image.extract_patches` (patch size 128×128 , stride 64), yielding a constant number of patches per image. When enabled, gender information is provided as an additional scalar input.

b) Training-only data augmentation.: During training, standard data augmentation operators are applied, including random horizontal flips, random rotations by multiples of 90°

($k \in \{0, 1, 2, 3\}$), brightness shifts ($\Delta \leq 0.08$), contrast scaling in $[0.92, 1.08]$, and additive Gaussian noise with standard deviation 0.02. Augmentation is applied exclusively to the training set; validation samples are never augmented.

c) Efficiency.: For computational efficiency, deterministically preprocessed tensors are cached, while stochastic augmentation is applied as a subsequent mapping during training, followed by batching and prefetching.

E. Training and validation protocol

Models are trained exclusively on the official training split and evaluated on the official validation split provided by the RSNA challenge. Since patient identifiers are not available, no patient-level splitting is performed. Instead, we rely on the predefined split, which preserves similar age and gender distributions across training and validation sets, as reported in the dataset documentation. This protocol matches the evaluation setting of the original RSNA Pediatric Bone Age Challenge [3] and ensures reproducibility.

V. LEARNING FRAMEWORK

A. Problem formulation

Let x denote a pediatric hand radiograph and $g \in \{0, 1\}$ a binary gender indicator. The objective is to learn a function $f_\theta(x, g)$ that predicts the corresponding bone age y in months. Rather than directly regressing a single scalar value, we formulate bone age estimation as the prediction of a discrete age distribution over months. This formulation allows the model to explicitly represent annotation uncertainty and ambiguous skeletal patterns. Model performance is evaluated using mean absolute error (MAE), measured in months, on the validation set.

B. Deep Label Distribution Learning

Bone age annotations are subject to inter-observer variability and intrinsic ambiguity, particularly in early developmental stages. To account for this uncertainty, we adopt Deep Label Distribution Learning (DLDL)[1]. Instead of regressing a single point estimate, the network learns to predict a soft probability distribution over the possible age range.

Let $y \in [0, 240]$ denote the ground-truth age in months and let $\mathcal{B} = \{0, 1, \dots, 240\}$ be the set of discrete age bins. During training, we dynamically generate a target distribution $\mathbf{q}$ for each sample using a Gaussian smoothing kernel centered at y with fixed standard deviation $\sigma = 4.0$:

$$q(i | y) = \frac{\exp\left(-\frac{(i-y)^2}{2\sigma^2}\right)}{\sum_{j \in \mathcal{B}} \exp\left(-\frac{(j-y)^2}{2\sigma^2}\right)}, \quad \forall i \in \mathcal{B}. \quad (2)$$

The standard deviation σ is a hyperparameter that controls label smoothing; we empirically set $\sigma = 4.0$ to capture reasonable annotation uncertainty.

The network outputs a logit vector $\mathbf{z} \in \mathbb{R}^{|\mathcal{B}|}$, which is transformed into a predictive distribution $\mathbf{p} = \text{softmax}(\mathbf{z})$. The loss function is the Kullback–Leibler divergence between

$\mathbf{q}$ and $\mathbf{p}$, implemented efficiently as the categorical cross-entropy with logits:

$$\mathcal{L}_{\text{DLDL}}(y, \mathbf{z}) = - \sum_{i \in \mathcal{B}} q(i | y) \log \left(\frac{\exp(z_i)}{\sum_j \exp(z_j)} \right). \quad (3)$$

a) *Expectation-based decoding.*: At evaluation time, a continuous age prediction is obtained by computing the expectation of the predicted distribution:

$$\hat{y} = \sum_{i \in \mathcal{B}} p(i) i. \quad (4)$$

Mean absolute error is then computed between $\hat{y}$ and the ground-truth age y .

C. Model architectures

CNN encoder. All models share a lightweight convolutional encoder trained entirely from scratch. The encoder consists of four blocks with increasing channel depths (32, 64, 128, 256). The first three blocks follow a VGG-style design: two stacked 3×3 convolutional layers with ReLU activation, followed by 2×2 max pooling and batch normalization. The fourth block contains a single 3×3 convolutional layer followed by max pooling and batch normalization. The final feature maps are spatially aggregated via Global Average Pooling (GAP), yielding a 256-dimensional feature vector $\mathbf{h}_{\text{img}}$.

Whole-image models (M1, M2). In the whole-image setting, the complete radiograph is processed by the shared CNN encoder. In model M2, gender metadata is embedded into a $d_g = 32$ dimensional vector $\mathbf{h}_{\text{gen}}$ using a fully connected layer with ReLU activation. The two representations are concatenated to form the joint embedding $\mathbf{z} = [\mathbf{h}_{\text{img}}; \mathbf{h}_{\text{gen}}]$.

This combined vector is processed by a projection head consisting of a fully connected layer with 256 units and ReLU activation, followed by Batch Normalization and Dropout ($p = 0.5$) to regularize the network. Finally, a linear layer maps the resulting features to $|\mathcal{B}|$ logits.

Patch-based models with attention (M3, M4). In the patch-based setting, each image is decomposed into a fixed grid of K overlapping patches. A shared CNN encoder processes each patch independently, yielding a set of patch embeddings $\{\mathbf{z}_k\}_{k=1}^K$.

Model M4 employs learned softmax fusion, in which a scalar importance score is predicted for each patch and normalized across patches:

$$s_k = \mathbf{w}^\top \mathbf{z}_k, \quad \alpha_k = \frac{\exp(s_k)}{\sum_{j=1}^K \exp(s_j)}, \quad \mathbf{z}_{\text{fused}} = \sum_{k=1}^K \alpha_k \mathbf{z}_k. \quad (5)$$

TABLE II
SUMMARY OF EVALUATED MODEL ARCHITECTURES.

Model	Input	Encoder	Fusion	Gender
M1	Whole image	Shared CNN	–	No
M2	Whole image	Shared CNN	–	Yes
M3	Patches	Shared CNN	Gated	Yes
M4	Patches	Shared CNN	Learned	Yes

Model M3 uses a gated attention mechanism to obtain more selective patch weighting:

$$\begin{aligned} \mathbf{h}_k &= \tanh(V \mathbf{z}_k) \odot \sigma(U \mathbf{z}_k), \\ s_k &= \mathbf{w}^\top \mathbf{h}_k, \\ \alpha_k &= \frac{\exp(s_k)}{\sum_{j=1}^K \exp(s_j)}, \\ \mathbf{z}_{\text{fused}} &= \sum_{k=1}^K \alpha_k \mathbf{z}_k. \end{aligned} \quad (6)$$

Here, V , U , and $\mathbf{w}$ are learnable parameters, $\sigma(\cdot)$ denotes the sigmoid function, and $\odot$ indicates element-wise multiplication.

In both patch-based models, the fused patch representation $\mathbf{z}_{\text{fused}}$ is concatenated with the gender embedding $\mathbf{h}_{\text{gen}}$ (if enabled) and passed through the same projection head (Dense 256 $\rightarrow$ BN $\rightarrow$ Dropout) as the whole-image model before the final prediction.

Table II summarizes the evaluated model architectures.

D. Training setup

All models are trained to predict age distributions using the DLDD loss described above. Optimization is performed using the AdamW optimizer with decoupled weight decay and a cosine learning-rate schedule. Batch normalization and dropout are applied in the prediction head to improve optimization stability and reduce overfitting. All architectural variants are trained under identical preprocessing, optimization, and early-stopping criteria to ensure a controlled and fair comparison across models.

VI. RESULTS

A. Evaluation metrics

Model performance is evaluated using mean absolute error (MAE), measured in months, on the official RSNA validation split. To explicitly analyze the accuracy–complexity trade-off, we additionally report the number of trainable parameters and the average inference time per image, measured using repeated forward passes with batch inference.

B. Main comparison at 256×256 resolution

Table III reports the primary results for all evaluated models trained and evaluated at an input resolution of 256×256 pixels. Model M1 serves as a whole-image baseline; M2 augments the image input with gender metadata; M3 and M4 evaluate patch-based models with gated and learned fusion mechanisms, respectively.

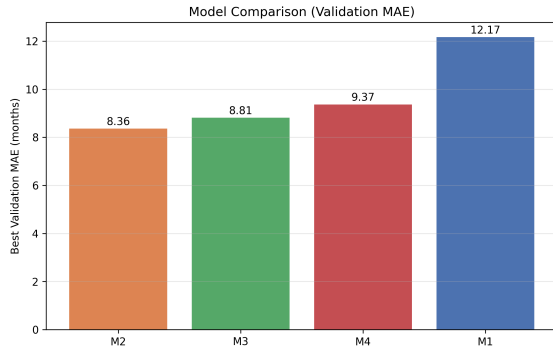


Fig. 5. Validation MAE for all evaluated models at 256×256 input resolution (lower is better).

TABLE III
COMPARISON OF EVALUATED MODELS ON THE RSNA VALIDATION SET
(256×256 INPUT).

Model	MAE (months)	Params (M)	Inference (ms/img)
M1: Image only	12.17	0.71	1.60
M2: Image + gender	8.36	0.72	1.72
M3: Patches + gated fusion + gender	8.81	0.79	12.83
M4: Patches + learned fusion + gender	9.37	0.72	12.77
Ensemble (M2 + M3 + M4)	7.80	–	–

Incorporating gender metadata yields a substantial improvement over the image-only baseline (M1→M2), reducing MAE by nearly four months. Patch-based models with attention-based fusion achieve competitive accuracy but do not outperform the whole-image model at equal resolution. Both gated and learned fusion substantially increase inference time (approximately $8\times$) due to multiple patch forward passes, making the accuracy–complexity trade-off explicit.

C. Training dynamics

Figure 6 shows the training and validation MAE curves for all models. While all models converge stably, patch-based variants exhibit higher run-to-run variability, reflecting the increased optimization difficulty and variance introduced by localized patch representations. Early stopping consistently selects an intermediate epoch that balances optimization progress and generalization.

D. Ensemble results

To further exploit the complementarity between whole-image and patch-based representations, we evaluated a probability-averaging ensemble combining the three strongest models at 256×256 resolution (M2, M3, and M4). Model outputs are combined at the probability level, and final age predictions are obtained via expectation-based decoding.

Using heuristic weights proportional to the inverse validation MAE of each individual model, the resulting ensemble achieves a validation MAE of **7.8 months**, improving upon all single-model results at the same resolution. Since ensemble weights are selected based on validation performance, this result may be slightly optimistic with respect to true generalization; nevertheless, it highlights the complementary nature

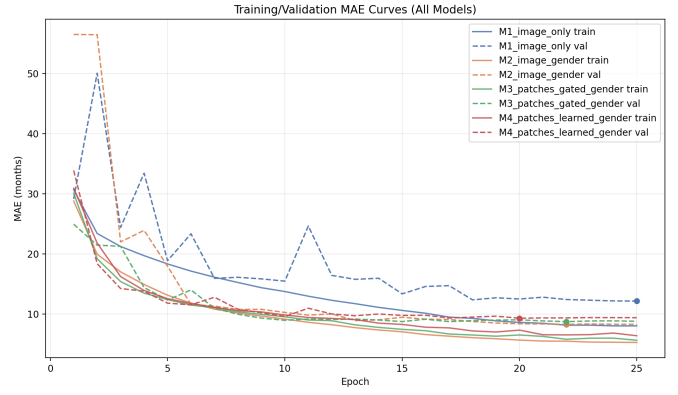


Fig. 6. Training and validation MAE curves for all evaluated models at 256×256 input resolution.

TABLE IV
EFFECT OF INPUT RESOLUTION ON MODEL M2.

Resolution	MAE (months)	Params (M)	Inference (ms/img)
256×256	8.36	0.72	1.72
512×512	7.15	2.56	3.54

of global and patch-level representations despite their differing computational profiles.

E. Resolution study

To assess the effect of input resolution independently of architectural changes, we evaluated the strongest whole-image model (M2) at a higher resolution of 512×512 pixels. Results are reported in Table IV.

Increasing the input resolution yields a significant accuracy gain (approximately 1.2 months MAE reduction) at the cost of higher computational complexity. Notably, the high-resolution whole-image model outperforms all patch-based variants while remaining computationally feasible, indicating that spatial resolution can be a stronger performance lever than patch decomposition under a train-from-scratch constraint.

F. Accuracy–complexity discussion

Taken together, the results highlight a clear and quantifiable accuracy–complexity trade-off. Gender metadata provides the largest single improvement with negligible computational overhead. Patch-based attention mechanisms offer interpretability and competitive accuracy but incur a substantial increase in inference time. Increasing input resolution yields the largest absolute accuracy gain, with a corresponding increase in memory footprint and latency. These findings emphasize the importance of jointly considering accuracy and computational cost when designing models under constrained training regimes.

G. Error analysis

To characterize failure modes beyond aggregate MAE, we analyze prediction errors of model M2. Figure 7 reports MAE stratified by gender, showing a small but consistent difference

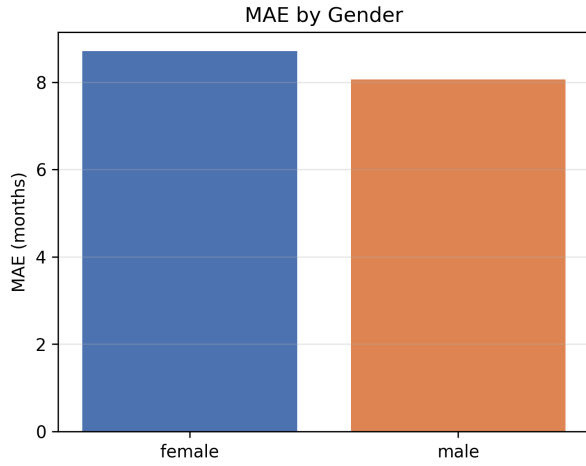


Fig. 7. Validation MAE stratified by gender for model M2.

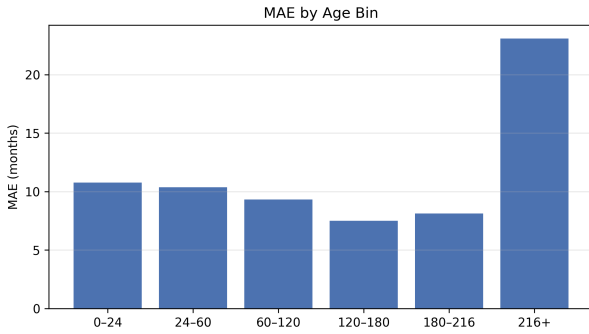


Fig. 8. Validation MAE across age ranges for model M2.

between groups. Figure 8 reports MAE across age ranges: errors are relatively stable across most pediatric age groups but increase noticeably at the upper end of the age spectrum (216+ months). This behavior is consistent with reduced visual distinction between late-adolescent skeletal structures, where small anatomical differences correspond to large changes in chronological age.

H. Patch-level interpretability

Figure 9 visualizes the learned patch weights of model M4. Each colored block corresponds to a patch in the fixed 3×3 grid; warmer colors indicate higher relative contribution to the final prediction, while cooler colors indicate lower contribution. Higher weights are consistently assigned to anatomically relevant regions such as the carpal bones and distal phalanges, providing coarse but faithful insight into the model’s decision process.

I. Qualitative results

To complement the quantitative evaluation, Figure 10 presents representative qualitative predictions from the validation set for model M4. The figure includes examples with low, typical, and high absolute error, together with ground-truth age, predicted age, absolute error, and gender.

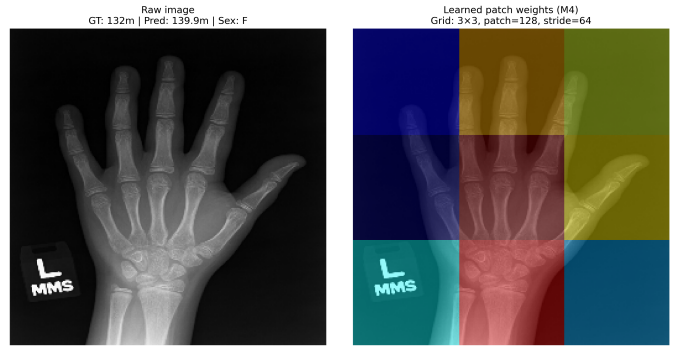


Fig. 9. Visualization of learned patch weights for model M4 (learned fusion). Images are shown in raw form for readability; weights are computed on preprocessed inputs.

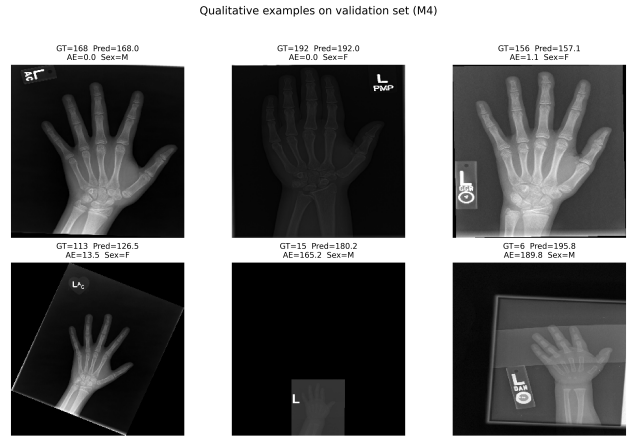


Fig. 10. Qualitative examples from the validation set for model M4. Images are shown in raw form for visualization, while predictions are obtained using the full preprocessing pipeline.

Correct predictions generally correspond to cases where ossification patterns of the wrist and phalanges are clearly visible, while larger errors often occur in younger patients or images with less distinctive skeletal structures. These qualitative observations are consistent with the error analysis and illustrate typical success and failure modes of the model.

VII. CONCLUDING REMARKS

This project investigated bone age estimation from pediatric hand radiographs under a strict train-from-scratch constraint, with the objective of improving predictive accuracy while maintaining methodological rigor. Building upon an initial baseline, we introduced a medically motivated preprocessing pipeline to mitigate shortcut learning from non-anatomical cues, adopted Deep Label Distribution Learning (DLDL) to explicitly model annotation uncertainty, and systematically evaluated multiple architectural strategies, including whole-image models, patch-based attention mechanisms, resolution scaling, and model ensembling.

Across all experiments, incorporating gender metadata consistently yielded the largest single improvement in accuracy at negligible computational cost, confirming its importance

as a complementary input for skeletal age assessment. Patch-based models with attention-based fusion achieved competitive performance and provided an interpretable mechanism for weighting local image regions, but did not outperform the strongest whole-image model at equal input resolution. Their primary benefit emerged when combined with global representations: a probability-level ensemble of whole-image and patch-based models reduced validation error beyond any individual model at the same resolution.

Increasing input resolution proved to be the most effective lever for improving single-model performance. The high-resolution whole-image model achieved the lowest validation error among all individual architectures, indicating that preserving fine-grained spatial detail can be more beneficial than decomposing the image into patches when training data and model capacity are sufficient. Importantly, this improvement was obtained while maintaining feasible inference latency, highlighting a favorable accuracy–complexity trade-off.

Overall, the results demonstrate that performance gains in bone age estimation arise from the interaction of multiple design choices rather than from a single architectural modification. Careful preprocessing, uncertainty-aware supervision, controlled architectural comparisons, and complementary model combination all contribute to improved robustness and accuracy under constrained training conditions.

A. Limitations

Despite these improvements, several limitations remain. First, patch extraction relies on a fixed overlapping grid without explicit anatomical localization, which may limit the effectiveness of patch-based models in the presence of large pose or scale variations. Second, error increases at the upper end of the age spectrum (216+ months), where skeletal structures exhibit limited visual progression despite chronological age changes. Third, ensemble weighting is determined heuristically using validation performance, which may introduce mild optimism in the reported ensemble results.

B. Lessons Learned and Difficulties

Lessons learned. This study highlights the importance of evaluating accuracy jointly with computational cost and of avoiding overly narrow conclusions based on single-model performance. In particular, we learned that higher spatial resolution and uncertainty-aware objectives can be more impactful than increasing architectural complexity alone. We also observed that training deep networks entirely from scratch requires careful attention to initialization and normalization, as standard recipes for pretrained models often fail in this regime.

Difficulties. We encountered specific implementation challenges, particularly regarding the patch-based pipeline. Managing tensor shape compatibility for the attention fusion module required several iterations to ensure correct gradient flow. Additionally, the strict train-from-scratch constraint imposed significant memory limitations; training the high-resolution (512×512) model required aggressive batch size reduction

to fit within GPU memory, which initially led to training instability before tuning the learning rate schedule.

Future work. Future work could explore anatomically informed region selection, lightweight localization modules, or age-dependent uncertainty modeling to further improve robustness, especially in underrepresented age ranges. Additional directions include leveraging unlabeled data through semi-supervised learning or incorporating longitudinal information when available.

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